

**B.Tech. Examination-2023**  
**Electronics & Communication Engineering Department**  
 (Odd Semester Regular and Supplementary)  
**Basic Electronics Engineering (ECEUGES01) for CSE**

Full Marks : 80

Time-3 hrs

(Answer any five)

(Symbols have their usual meaning)

1. (a) Explain p-type semiconductor. Draw the majority and minority carrier concentration curve of it. 3+2  
 (b) Establish the following relation between drift velocity, mobility and electric field for an electron in semiconductor under thermal equilibrium :-

$$v = \mu_n E, \text{ where } \mu_n = \frac{q\hbar}{m_n^*} \quad 8$$

- (c) Draw the energy-band diagram of n-type semiconductor. 3

2. (a) Write the Fermi-Dirac distribution function. Define Fermi-level. 4+3  
 (b) Show that the Fermi-Dirac distribution function is rectangular at 0K. 6  
 (c) Write down the diffusion current expression of electron. 3

3. (a) Show that equilibrium concentrations of electrons ( $n_0$ ) and holes ( $p_0$ ) in a semiconductor can be expressed as :-

$$(i) \quad n_0 = N_c e^{-\left(\frac{E_c - E_F}{kT}\right)} \quad (ii) \quad p_0 = N_v e^{-\left(\frac{E_F - E_v}{kT}\right)} \quad 5+5$$

- (b) Show that the product of electron and hole concentrations under equilibrium is constant and can be expressed as :-  $n_0 p_0 = n_i^2$ . 6

4. (a) Explain the formation of depletion region and barrier potential  $V_0$  in a p-n junction diode under equilibrium. 4+4  
 (b) Explain with diagram the forward bias and reverse bias characteristics of p-n junction diode. 4+4
5. Determine the diode dynamic resistance ' $r_d$ ' and cut-in voltage ' $V_f$ ' of the forward-bias diode at room temperature with the following parameters :-

$$I_S = 1 \text{ nA}, \quad \eta = 2, \quad \text{near the region where the diode current is } 3 \text{ mA}. \quad 8+8$$

6. (a) Explain the operation of the clipper circuit with the input shown in Fig. 1, assuming the diode as ideal diode. 5  
 (b) Find out the changes in output voltage waveform if the diode is a real one with cut-in voltage  $V_f$  and diode dynamic resistance  $r_d$  after estimating the output voltage expression. 6  
 (c) Draw and explain the output of the clamper circuit shown in Fig. 2 with the given sinusoidal input voltage. 5

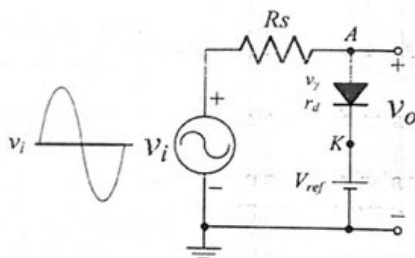


Fig. 1

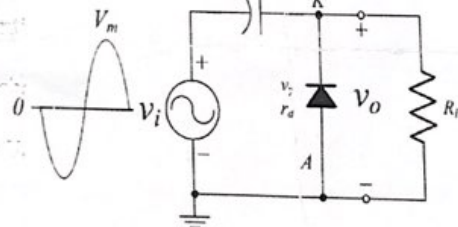


Fig. 2

7. (a) What are the ideal characteristics of OPAMP ? 3  
 (b) Determine the gain expression of non-inverting amplifier. 5  
 (c) What is an unity-gain buffer ? 3  
 (d) Determine the output voltage in an inverting input configuration of OPAMP with  $R_1 = 1k\Omega$ ,  $R_2 = 2k\Omega$  and input voltage  $v_i = 3V$ . 5
8. (a) Explain the operation of full wave rectifier circuit using bridge network. 3+4  
 (b) Show that the output DC voltage ( $V_{dc}$ ) and ripple factor ( $\Gamma$ ) of full wave rectifier output is  
 (i)  $V_{dc} = \frac{2V_m}{\pi}$  (ii)  $\Gamma = 48.3 \%$  3+(3+3)
-

**ARABIC**

**(USE SEPARATE ANSWER SHEET FOR ARABIC)**

**Q. 1. Attempt any ten, each question carries equal marks**

**2×10=20**

i. Merge the following letters and make a word:

س + ب + ي + ل ، ن + ظ + ي + ف

ii. Identify the letters and Split the following words: يستبشر، صالحين

iii. Read and write the following words in English: لُخْرِجَتْهُمْ، لُبِّيْنَةُ

iv. Identify the following words either 'Noun' (اسم) or 'Verb' (فعل) :

عَالِيَّة، سَوْفَ يَكْتُبُ، الْجَامِعَةُ، قَدْ أَفْلَحَ

v. Change the Gender of following words:

معلمة، طالب، أكبر، قانِئ.

vi. Identify the following words either definite (المعرفة) or indefinite (النكرة) :

المُعَلِّم طَبِيبٌ يَا زَكِي خَيْرٌ

vii. Write down the Example of Huruf (الحروف) from parts of speech.

viii. Write any four demonstrative Pronouns (اسم الإشارة) of Arabic.

ix. How many kinds of sentence are there in Arabic?

x. How many kinds of Madd are in Arabic? and what are they?

xi. What do you Know about Shaddah ? write in short with example.

xii. Change the Nominal sentence into verbal sentence and Verbal sentence

into Nominal Sentence. رَجَعَ عُمَرُ مِنَ الْمَدْرَسَةِ الْوَلَدُ دَخَلَ فِي

الفصل

**Q. 2. Attempt any 5 Questions, each question carries equal marks.**

**5×4=20**

i. Write the Arabic numbers 1-5 in Arabic scripts.

ii. Write down the any five signs of الفعل (Verb) from the parts of speech (أقسام الكلام) by citing of example.

iii. Write down the Moon Letters (الحروف القمرية) and Sun Letters (الحروف الشمسية) in Arabic.

iv. How to identify Definite Noun (المعرفة) Write any five with example.

v. What do you know about Nominal Sentence (الجملة الاسمية) and Verbal Sentence (الجملة الفعلية) Write down with example.

vi. Write down the signs of femininity (علامة التأنيث) with proper examples.

Examinations 2022-2023

Autumn (Odd) Semester

UG Programme

Course Code : VCCUGAU01

Course Title: Compulsory Islamic Studies

Answer Any Four Question :

5x4=20

X 2 = 40

1. What is the meaning of Tawheed and its types? Describe it.
2. Describe the life of the Prophet Muhammad (PBUH).
3. Discuss about the five daily prayers.
4. What is the Concept of "Risalah" (Prophethood) in Islam? Explain it.
5. Describe the symbol of "Qiyamat." (Day of Judgement)
6. Write a short note about Roja (Fasting).



ALIAH UNIVERSITY  
END SEMESTER EXAMINATION (AUTUMN) - 2023  
COURSE CODE: MATUGBS01  
COURSE NAME: ENGINEERING MATHEMATICS I  
(FOR ECE, EEN, CSE, BCA, MEN, CEN)

Group A (40 Marks)

1. Answer the following Questions: 1X5=5

- (a) Define a sequence of real numbers.
- (b) Show that the series  $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$  is convergent and find its sum.
- (c) Define the radius of convergence of a power series.
- (d) Define a Beta function.
- (e) If  $\Gamma(n)$  ( $n > 0$ ) denotes the Gamma function, then show that  $\Gamma(n+1) = n\Gamma(n)$ .

2. Answer any one of the following questions: 5X1=5

- (a) If  $x_n = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)}$ , then show that the sequence  $\{x_n\}$  is bounded and monotonically increasing. Also find the limit of the sequence  $\{x_n\}$ .
- (b) If  $x_n = \frac{4n+3}{3n+4}$ , then show that the sequence  $\{x_n\}$  is bounded above and monotonically increasing. Find limit of the sequence  $\{x_n\}$ .

3. Answer any one of the following questions: 5X1=5

- (a) Test the convergence of the following series:  
$$1 + \frac{1}{2} \cdot \frac{1}{3} + \frac{1.3}{2.4} \cdot \frac{1}{5} + \frac{1.3.5}{2.4.6} \cdot \frac{1}{7} + \dots$$
- (b) Use Cauchy's root test to examine the convergence of the series  $\sum u_n$ ,  
where  $u_n = \left\{ \left( \frac{n+1}{n} \right)^{n+1} - \frac{n+1}{n} \right\}^{-n} \cdot \frac{1}{e-1}$

4. Answer any one of the following questions: 5X1=5

- (a) Evaluate  $\lim_{x \rightarrow 0} \frac{x(5^x - 1)}{1 - \cos x}$ .
- (b) If  $f(x) = \begin{cases} x+1, & \text{when } x \leq 1, \\ 3-ax^2, & \text{when } x > 1, \end{cases}$   
then for what value of  $a$ , the function  $f(x)$  will be continuous at  $x = 1$ .

5. Answer any one of the following questions: 5X1=5

- (a) If  $y = e^{ax} \sin bx$ , then show that  $y_n = (a^2 + b^2)^{\frac{n}{2}} e^{ax} \sin \left( bx + n \tan^{-1} \frac{b}{a} \right)$ ,  
where  $y_n = \frac{d^n y}{dx^n}$ .
- (b) If  $y = \frac{1}{x^2 + a^2}$ , then show that  $y_n = \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1} \theta \sin(n+1)\theta$ ,  
where  $y_n = \frac{d^n y}{dx^n}$ .

6. Answer any one of the following questions: 5X1=5

- (a) If  $y = \tan^{-1} x$ , then prove the following:

(i)  $(1+x^2)y_2 + 2xy_1 = 0$ ,

(ii)  $(1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$ .

(b) If  $2x = y^{\frac{1}{m}} + y^{-\frac{1}{m}}$ , then show that  $(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0$ .

7. Answer any one of the following questions:

5X1=5

(a) State and prove Lagrange's Mean Value theorem.

(b) Using L' Hospital's rule evaluate  $\lim_{x \rightarrow 0} \left( \frac{\sin x}{x} \right)^{\frac{1}{x}}$ .

8. Answer any one of the following questions:

5X1=5

(a) If  $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ , then show that  $I_n = \frac{n-1}{n} I_{n-2}$ , and hence find the value of  $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$ .

(b) Show that  $\int_0^{\infty} e^{-x^4} x^2 \, dx \times \int_0^{\infty} e^{-x^4} \, dx = \frac{\pi}{8\sqrt{2}}$ .

Group B (40 Marks)

Answer any four questions.

1.a) Transform the equation  $\sqrt{r} = \sqrt{a} \sin(\theta/2)$  to its Cartesian form.

b) Find the equation of the three planes passing through the points  $(3,1,1)$ ,  $(1,-2,3)$  parallel to the co-ordinate axes. (5+5)

2.a) Evaluate  $\int_C f(x,y) \, dx$  where  $f(x,y) = x^2 + y^3$  and the curve  $C$  is the arc of the parabola  $y = x^2$  in the  $xy$ -plane from  $(0,0)$  to  $(1,1)$ .

b) Find the equation of the cone whose vertex is the origin and which passes through the curve of intersection of the plane  $lx + my + nz = p$  and the surface  $ax^2 + by^2 + cz^2 = 1$ . (5+5)

3.a) Evaluate  $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2 + y^2) \, dy \, dx$  by changing to polar co-ordinates, with the help of Jacobian.

b) If the co-ordinates of  $A, B, C, D$  are  $(-1, -2)$ ,  $(7, 4)$ ,  $(4, 8)$  and  $(-4, 2)$  respectively, then show that  $ABCD$  is a rectangle. (7+3)

4. Define limit and continuity of a vector function. If derivative of  $F(t)$  exist at  $t = t_0$ , then prove that  $F(t)$  is continuous at  $t = t_0$ . Is the converse true? Justify your answer. (4+3+3)

5.a) Evaluate  $\int_0^a \int_0^x \int_0^y x^3 y^2 z \, dz \, dy \, dx$ .

b) Evaluate  $\int_0^{\pi/2} \int_0^{\pi} \sin(x+y) \, dy \, dx$ . (7+3)

6. a) Find the volume of the tetrahedron formed by the four planes  $lx + my + nz = p$ ,  $lx + my = 0$ ,  $my + nz = 0$  and  $nz + lx = 0$ .

b) State Serret-Frenet's Formulae. (7+3)

7. State Green's theorem for a plane. Verify the theorem for  $\oint_C [(xy + y^2) \, dx + x^2 \, dy]$  where  $C$  is the closed curve of the region bounded by  $y = x$  and  $y = x^2$ . (3+7)



**End Term Examination of Autumn Semester 2022**  
(For 1<sup>st</sup> Year 1<sup>st</sup> Semester of 4-Year B.Tech)

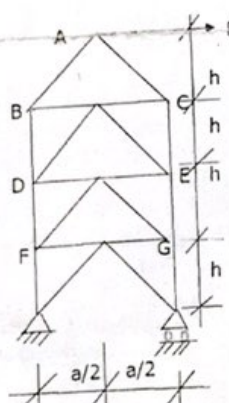
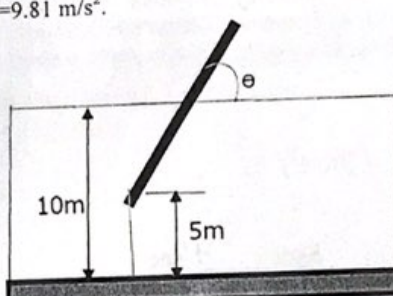
Subject: **ENGINEERING MECHANICS**  
Full Marks: 80

Paper Code: **MENUGES01**  
Time: 3 hours

**Group- A.**

(Answer any five questions)

[5x (3+5) =40]

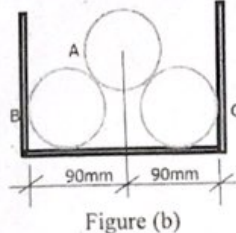
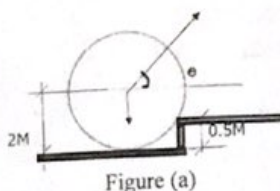
- |                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. (a) State the polygon law (vector). Why current is not treated as vector quantity?<br/>(b) A rigid body is spinning with an angular velocity of 3rad/sec about an axis parallel to <math>2i+j-2k</math> passing through the point <math>i+3j-k</math>. Find the velocity of the point of the body whose position vector is <math>4i+8j+k</math>.</p>                               | <p>2. (a) State the principle of resolution of forces. What is the moment of couple?<br/>(b) The forces 20N, 30N, 40N, 50N and 60N are acting at one angular point of regular hexagon, towards the other five points, taken in order. Find the magnitude and direction of the resultant.</p>                                                                                                                                                                                                                |
| <p>3. (a) State the principle of equilibrium of a body and equilibrium forces.<br/>(b) Draw the F.B.D of following figure and find the reaction on wall &amp; tension in the rope. Length of rope is 10cm, the angle between rope and wall is <math>30^\circ</math> &amp; weight of ball is 100 N.</p>  | <p>4. (a) What is friction force? State three Coulomb's Law for dry friction?<br/>(b) A uniform ladder of weight 250N rests against a smooth vertical wall and a rough horizontal floor making an angle of <math>45^\circ</math> with horizontal. Find the force of friction at the floor using method of virtual works.</p> <p>6. (a) Define Mass Moment of Inertia &amp; radius of Gyration.<br/>(b) Find Mass moment of Inertia of cylinder having height "h" &amp; radius "R".</p>                      |
| <p>5. (a) What are the assumptions of truss? State the differences between truss and frame.<br/>(b) Find the axial force in members DF and EG in the given figure.</p>                                                                                                                                | <p>7. (a) Define static fluid. What will be the shear force if a fluid is in rest?<br/>(b) A buoy in the form of a uniform 8m pole 0.2m dia has a mass of 200kg and is secured at its lower end to the bottom of a fresh water lake with 5m of cable. If the depth of the water is 10m, calculate the angle <math>\theta</math>, made by the pole to the horizontal. Assume "g" = <math>9.81 \text{ m/s}^2</math>.</p>  |

**Group-B**

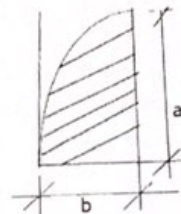
(Answer four questions only)

[4x (10) =40]

1. (a) Find Minimum force "P" to just start the roller as shown in figure (a). Given Weight of Roller is 100N.  
(b) Three Cylinders weighing 100N each and of 80mm dia are placed in a channel of 180mm width as shown in figure (b). Determine the pressure exerted by (i) the cylinder A on B at the point of contact, (ii) the cylinder on the wall. [3+7]



2. (a) One solid and one hollow cylinder having same mass are kept on two inclined planes with same inclination and same height from the ground. Now both the cylinders are freed to move on the planes. Which cylinder will reach the ground first? Justify your reason.  
(b) Find centroid ( $x_c$  &  $y_c$ ) of the parabola  $x=ky^2$  as shown in figure. [2+8]

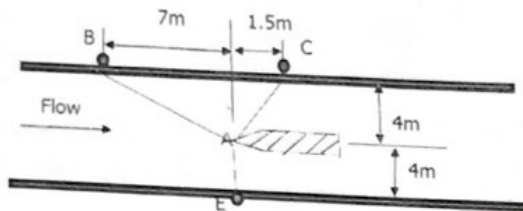




3. (a) State Lami's theorem.

(b) For a new sailboat, a designer wants to determine the drag force that may be expected at a given speed. To do so she places a model of the proposed hull in a test channel and uses three cables to keep its bow on the centerline of the channel. Dynamometer readings indicate that for a given speed, the tension is 200N in cable AB and 300N in cable AE. Determine the drag force exerted on the hull and the tension in cable AC.

[2+8]



6. (a) A car starts from rest with an acceleration of  $4\text{ m/s}^2$ . What is the distance travelled by the car in 8th sec of its travel?

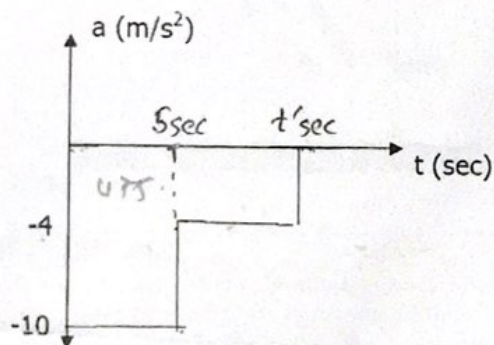
(b) Car A starts from rest at  $t=0$  and travels along straight road with a constant acceleration of  $2\text{ m/s}^2$  until it reaches a speed of  $25\text{ m/s}$ . Afterwards it maintains this speed. Also, when  $t=0$ , car B located at  $2000\text{ m}$  down the road is travelling towards car A at a constant speed of  $20\text{ m/s}$ . Determine the distance travelled car A when they pass each other.

[3+7]

7. (a) The acceleration of a particle is given by  $a=10/(v+1)$ , where  $a$  is the acceleration in  $\text{m/s}^2$  and  $v$  is the velocity in  $\text{m/s}$ . The particle starts with zero initial velocity. Find the distance (in m) in which the particle will attain a velocity of  $10\text{ m/s}$ .

(b) An aeroplane travelling at  $70\text{ m/s}$  lands on a straight runway and has a deceleration described by the following graph. Determine the time,  $t$ , in sec to reach a speed of  $5\text{ m/s}$ .

[5+5]



4. (a) A hollow steel tube  $3.5\text{ m}$  long has external diameter  $120\text{ mm}$ . In order to determine the internal dia, the tube was subjected to tensile load of  $400\text{ kN}$  and its extension was measured to be  $2\text{ mm}$ . If the modulus of elasticity for the tube material is  $200\text{ GPa}$ , determine the internal dia of the tube.

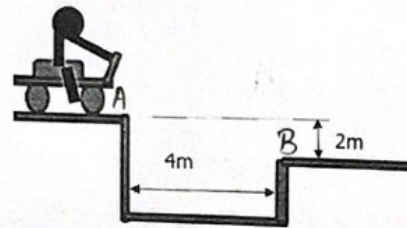
(b) A composite shaft is made by enclosing aluminium rod of  $60\text{ mm}$  dia, in brass tube of  $10\text{ mm}$  thickness. A composite shaft is subjected to a Torque of  $5\text{ kN-m}$ , find the maximum shear stress in both the shafts (Given Modulus of Rigidity,  $G=28\text{ GPa}$  &  $40\text{ GPa}$  for Aluminium and Brass Respectively.

[5+5]

5. (a) A gun fires two bullets at  $300$  and  $600$  with horizontal. They have the same range. The ratio of their maximum height will be \_\_\_\_\_?

(b) A motorcyclist wants to jump over a ditch as shown. Find the minimum velocity necessary at point A in  $\text{km/hr}$ . Also find inclination and magnitude of the velocity of the motor cycle just after clearing the ditch.

[3+7]



8. (a) A block of mass  $5\text{ kg}$  resting on a  $30^\circ$  inclined plane is released from rest. After travelling a distance of  $0.8\text{ m}$  along the plane, the block hits a spring of stiffness  $1\text{ m/mm}$ . Find the maximum compression of the spring (in mm). Assume co-eff of friction  $=0.2$  and  $g=10\text{ m/s}^2$ . Use figure (a).

(b) Find the acceleration of a solid right circular roller "A" of mass  $20\text{ kg}$  when it is being pulled by another body "B" of mass  $10\text{ kg}$  along a horizontal plane as shown in figure. The mass "B" is attached to the end of a string wound round the circumference of the roller. Assume that there is no slipping of the roller, the string is inextensible and Pulley is frictionless. Use figure (b).

[5+5]

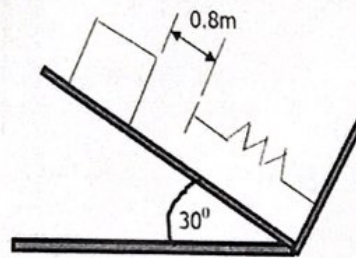


Figure (a)

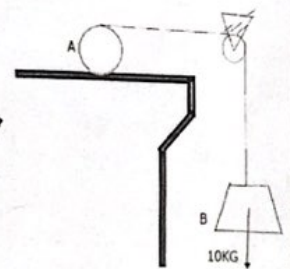


Figure (b)